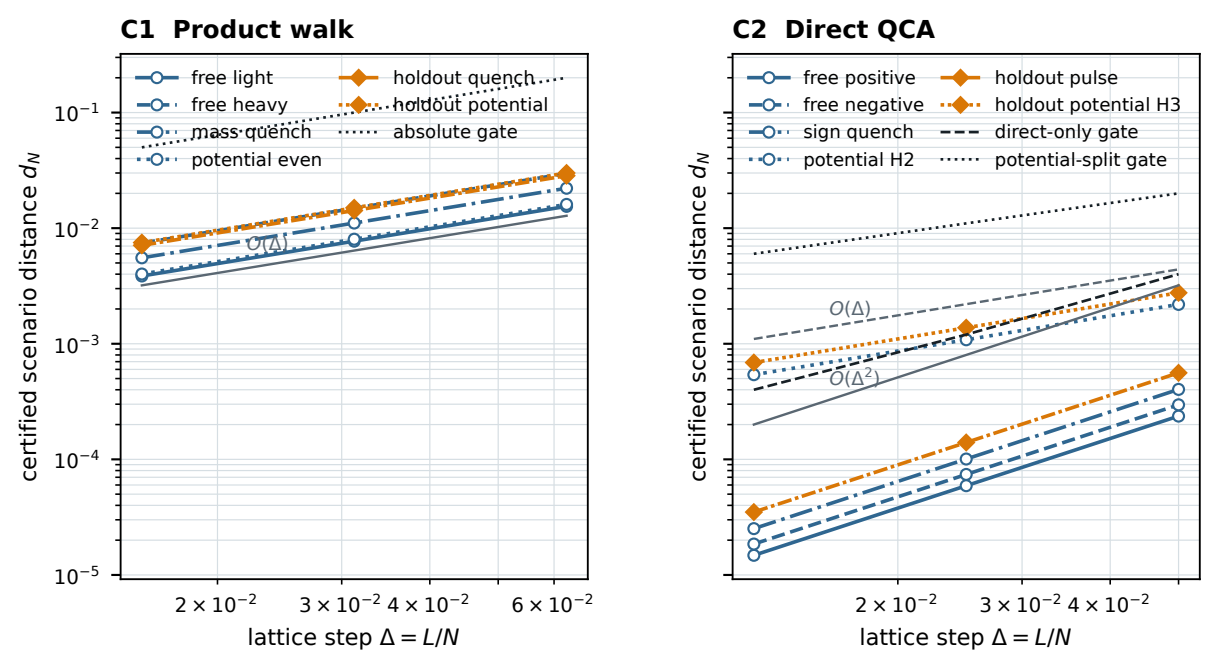
C1 Product walk
free light
free heavy
mass quench
potential even
holdout quench
holdout potential
absolute gate
$O(\Delta)$
certified scenario distance d_N
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
lattice step $\Delta = L/N$
C2 Direct QCA
free positive
free negative
sign quench
potential H2
holdout pulse
holdout potential H3
direct-only gate
potential-split gate
$O(\Delta)$
$O(\Delta^2)$
certified scenario distance d_N
2×10^{-2}
3×10^{-2}
4×10^{-2}
lattice step $\Delta = L/N$